Eddie Ripple

Natrona Heights, PA
www.ednotes.wiki/personal

Email: edrip222@gmail.com
Phone: 724 - 448 - 9767
Github: erip3

EDUCATION

The Pennsylvania State University

- BS in Computer Science — 2022-2026 GPA: 3.98
  Minors in Cybersecurity and Computer Engineering

Relevant Coursework

Database Management, Systems Programming, Computer Architecture, Deep Learning, Data Structures and Algorithms, Computer Vision, Web Apps with OOP, Computer Security

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TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, Verilog, Embedded C
Frameworks & Libraries: Spring Boot, React, Vite, Pandas, NumPy, Matplotlib
Data & Databases: PostgreSQL, MySQL, SQLite, Data Analysis
Tools & Infrastructure: Docker, Git, GitHub Actions, DigitalOcean, Linux/Unix, VS Code

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RELEVANT EXPERIENCE

EdNotes.wiki — *Summer 2026*

- Developed a full-stack web application serving as both a personal portfolio and an interactive platform for publishing computing articles and demonstrations.
  - Built a responsive frontend using **React & TypeScript** with a **Java (Spring Boot)** backend and **PostgreSQL** database.
  - Containerized the backend with **Docker** and deployed to **DigitalOcean.**

Simulated Thread Scheduler — *Fall 2025*

- Developed a low-level **C** program using the **pthreads** library to simulate an operating system's thread scheduling.
  - Implemented a modular scheduling interface supporting First-In-First-Out (FIFO), Shortest Remaining Time First (SRTF), and Multi-Level Feedback Queue (MLFQ) algorithms.

IT Auditor Intern: *ATI - Pittsburgh, PA* — *Summer 2025*

- Built a data analysis tool to identify duplicate invoices and erroneous payments, detecting duplicates with a combined worth of ~$100,000.
  - Wrote tests with **Pandas** and **SQL** to flag potential duplicates.
  - Designed a user interface using **HTML** and **CSS**, documented code, and wrote a user manual to support adoption and ease of use.

Embedded Systems Fan Control Project — *Spring 2024*

- Designed and implemented a fan control system using both an **FPGA (Verilog)** and an **Arduino Microcontroller (Embedded C)** for comparative evaluation.
  - Analyzed tradeoffs between FPGA-based and microcontroller-based approaches.
  - Presented findings at Penn State New Kensington Research Expo.

EXTRACURRICULARS

Volunteer PowerPoint Creator — 06/2022 - Present
Central Presbyterian Church, Tarentum, PA

Tau Beta Pi — Inducted 08/2024
University Park, PA

Penn State New Kensington Honors Program — 08/2022 - 05/2024